

# Switching the loop on at inference

## cuts E rate $\approx 15\times$

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The trained networks this entry uses are produced once in the shared training hub, exp022 (Training), and reused here rather than retrained.

### Abstract

Loads a trained COBA network and switches the I-loop on at inference by sweeping  $ei\_strength$  from 0 to 1. The same feedforward weights that fire at  $\approx 133$  Hz without the loop fire at  $\approx 9$  Hz with it engaged, a  $\approx 15\times$  drop with no weight update. Accuracy, though, falls  $\approx 36$  pp (from  $\approx 90\%$  to  $\approx 55\%$ ) as the loop engages, because the readout was never trained with it. PING gating is a post-hoc sparsity knob: the architecture, not the training, supplies the gamma dynamics, but using it well still needs training *with* the loop.

### Method

| Parameter                                      | Value                                                                   |
|------------------------------------------------|-------------------------------------------------------------------------|
| Integration timestep $\Delta t$                | 0.1 ms                                                                  |
| Trial duration $T$                             | 200 ms                                                                  |
| MNIST samples (80/20 stratified split of 7000) | 5600 train / 1400 test ( $\approx 10\%$ of the 70k-sample MNIST corpus) |
| Epochs                                         | 50                                                                      |

The PING and COBA baseline definitions and the training recipe are in exp025 this entry runs the COBA  $\rightarrow$  PING I-loop transfer probe at eval time on the trained baselines. (The input-rate sweep / f-I curve material that previously lived here has moved to exp023, the natural home for “architectural response to drive”.)

**Inference-time probe.** The trained COBA baseline (seed 42,  $\theta_u = \text{off}$ ) is loaded and  $ei\_strength$  (the I-loop gain) is overridden at eval time across 11 values from 0 to 1.  $W_{\text{in}}$  and  $W_{\text{out}}$  load from the COBA checkpoint;  $W^{EI}$  and  $W^{IE}$  are freshly initialised (the COBA checkpoint stores these at zero, so skipping the load leaves a functional I-loop). No retraining.

## Results

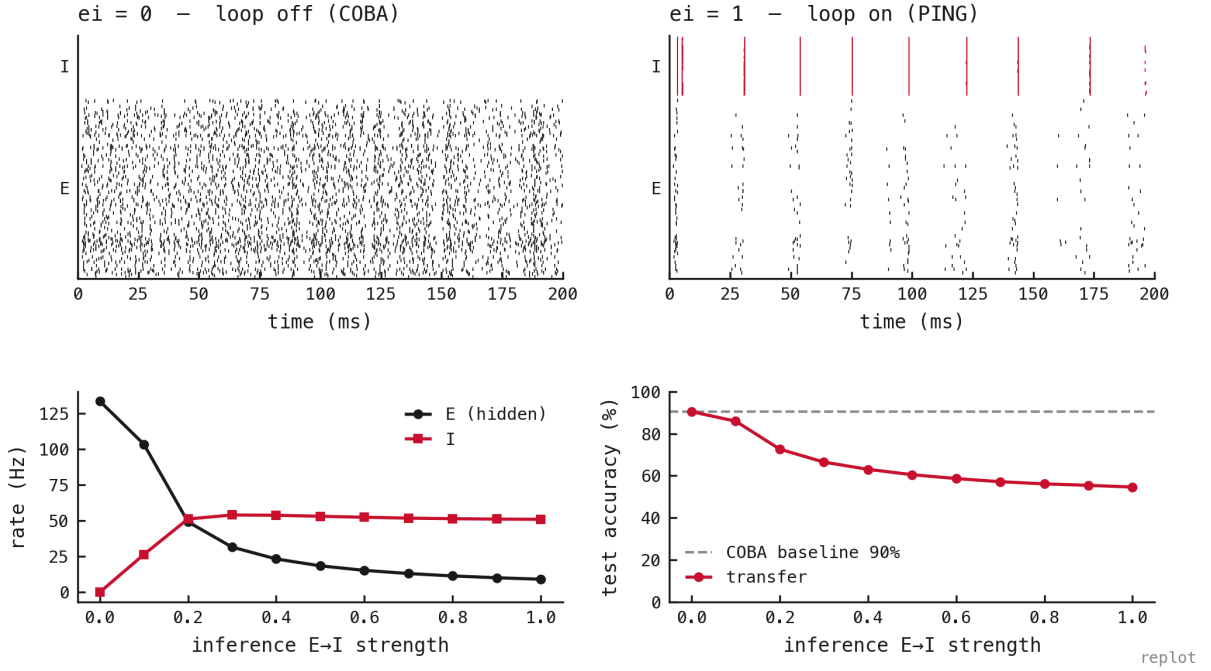


Figure 1: Switching the recurrent I-loop on *at inference* on a trained COBA network (no retraining;  $W^{EI}/W^{IE}$  freshly wired, since the COBA checkpoint stores them at zero). **Top:** the same feedforward weights fire densely and asynchronously at  $ei = 0$  (COBA) and in gamma bands at  $ei = 1$  (PING); the gamma dynamics come from the inhibitory architecture, not from training. **Bottom left:** E rate falls  $\approx 15\times$  ( $\approx 133 \rightarrow 9$  Hz) as the loop engages while I rises to  $\approx 51$  Hz; the suppression is continuous in loop strength. **Bottom right:** accuracy *degrades* without retraining, from the  $\approx 90\%$  COBA baseline to  $\approx 55\%$  at full strength (a  $\approx 36$  pp cost). So the architecture supplies the rate-gating for free, but using it well needs training *with* the loop (exp025): the sparsity is architectural, the accuracy is learned.

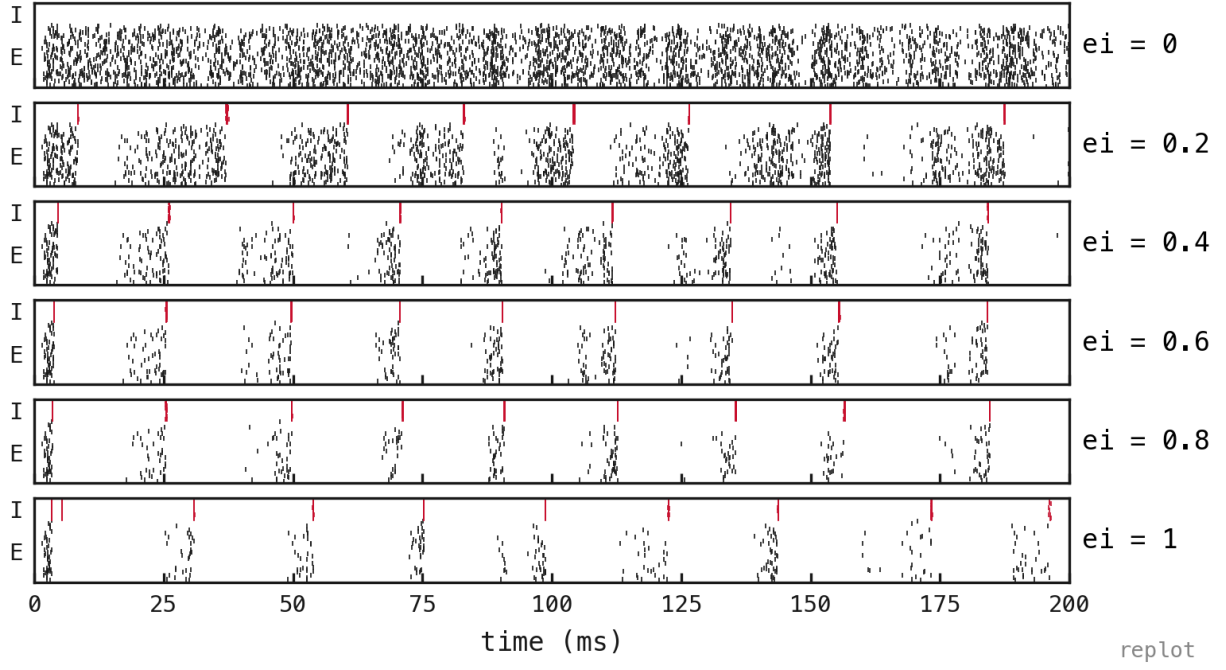


Figure 2: The full transition: trained COBA replayed at six inference-time  $ei\_strength$  values (same trial, same feedforward weights, a fresh I-loop each row). At  $ei = 0$  the asynchronous-dense COBA pattern persists; by  $ei \approx 0.4$  the same weights produce gamma cycles, sharpening toward  $ei = 1$ . The rhythm appears continuously as the loop is wired in, with no retraining at any point.